

Generation Interconnection System Impact Study Report

For Queue Project Q23-2315

Trans Bay Cable Facility 138 kV

May 26, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2315. The Interconnection Customer proposes a 87.2 MW solar plus storage facility interconnecting at the Trans Bay Cable Facility 138 kV substation in the Virginia area. The request was received on May 22, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 408 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$305,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ruritan – Gosport 138 kV line	Victory Way – Wheelabrator Portsmouth 138 kV L	108.9	230	13.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ruritan – Gosport 138 kV line	Reconductor line	\$2,661,000	12% (\$305,000)
	Total allocated to project		\$305,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.